

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Title – Nutrition Agent

Domain of Project – Health & Nutrition

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GitHub: github.com/DevSatyam7

Problem Statement -

Problem Statement : Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with **diet planning, portion control, and tracking nutritional intake**. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an **AI-powered Nutrition Agent** that delivers **personalized dietary insights and guidance** for individuals, families, and health professionals. The solution should:

- **Personalized Diet Plans** – Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
- **Food Data RAG Retrieval** – Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.
- **Diet Tracking & Feedback** – Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
- **Preventive Health Focus** – Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).
- **Multi-Agent System** –
 - **Nutrition Knowledge Agent** (fetches nutritional data)
 - **Diet Recommendation Agent** (personalized meal plans)
 - **Health Advisory Agent** (preventive health suggestions)
 - **Food Log & Feedback Agent** (meal tracking + analysis)
- **Visualization Dashboard** – Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent

The proposed solution is a **multi-agent AI-powered Nutrition Agent** built using **IBM Bob**, **IBM watsonx.ai**, and **Granite Models** to provide personalized, real-time dietary guidance. The system addresses challenges in diet planning, nutrition tracking, and preventive healthcare through an intelligent, modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate nutritional data from trusted sources such as USDA and WHO datasets. This data is embedded and stored in a vector database, enabling the system to retrieve and summarize food nutrition details (calories, macros, vitamins) using Granite models.

The system consists of four key agents:

Nutrition Knowledge Agent retrieves and summarizes nutritional information using RAG.

Diet Recommendation Agent generates personalized meal plans based on user inputs such as age, health conditions, dietary preferences, and fitness goals.

Health Advisory Agent provides preventive health suggestions, including disease-specific diets (e.g., diabetes-friendly or heart-healthy plans).

Food Log & Feedback Agent allows users to log meals via text, voice, or images and delivers instant nutritional analysis with actionable feedback.

The solution includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes daily and weekly nutrient intake, highlights deficiencies, and tracks progress toward health goals.

IBM Bob is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent nutrition assistant** that promotes healthier lifestyles through personalized insights, real-time feedback, and preventive healthcare recommendations.

Technology Used

- **IBM Bob platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG** – RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Required Files created using IBM bob -

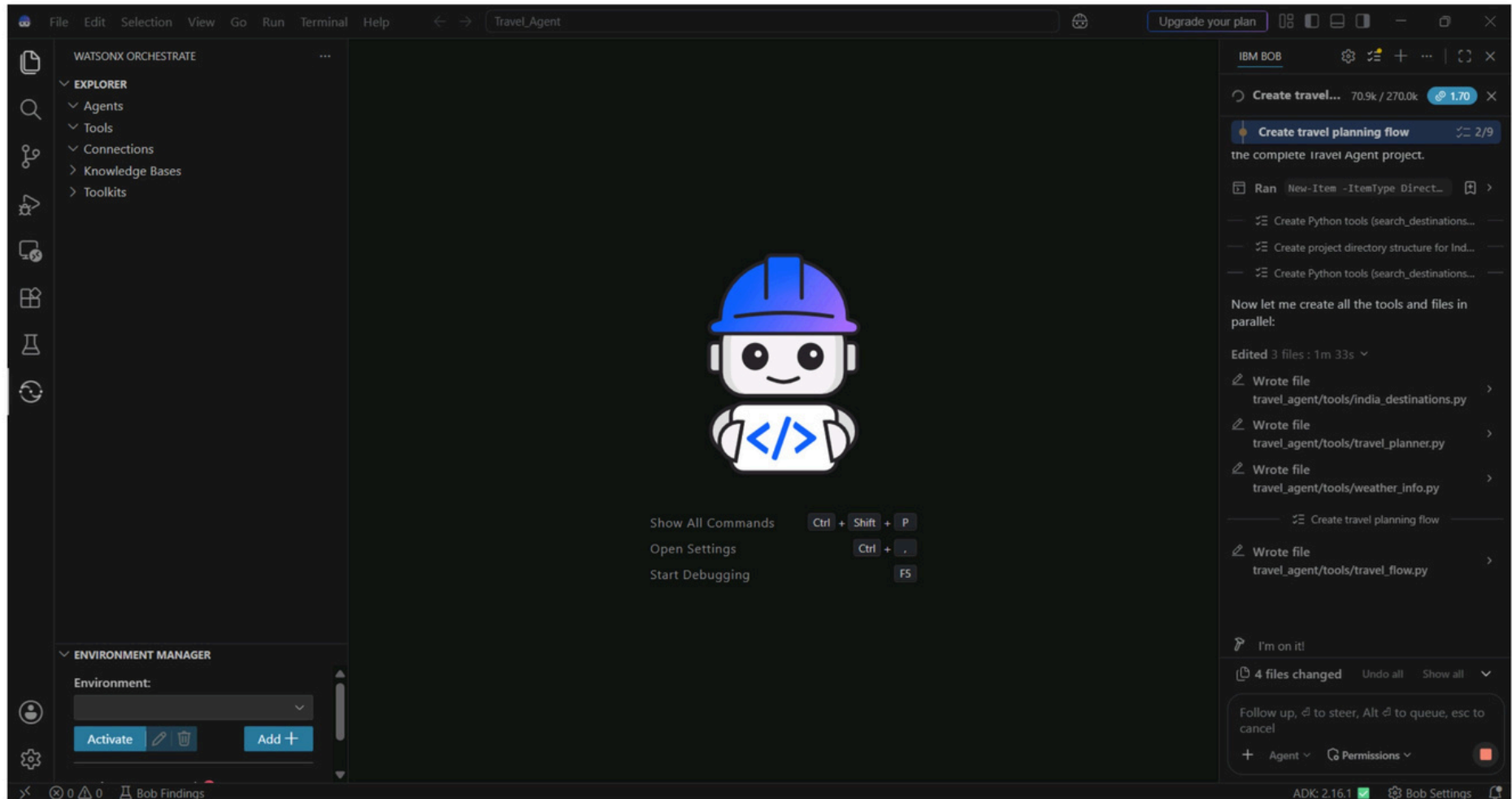
The screenshot displays the IBM WatsonX Orchestrate interface. The main editor shows a Python file named `travel_flow.py` with the following code:

```
1 from pydantic import BaseModel, Field
2 from typing import Optional
3 from ibm_watsonx_orchestrate.agent_builder.tools import tool, ToolPermission
4 from ibm_watsonx_orchestrate.flow_builder.flows import Flow, flow, START, END
5
6
7 # -- Input / Output Schemas --
8
9 class TravelFlowInput(BaseModel):
10     """Input schema for the India travel planning flow."""
11     destination: str = Field(description="Destination in India (e.g., Goa, Jaipur, Kerala Backwater)", ge=1, le=30)
12     duration_days: int = Field(description="Number of days for the trip (1-30)", ge=1, le=30)
13     num_travelers: int = Field(default=2, description="Number of travelers", ge=1, le=20)
14     month: str = Field(description="Month of travel (e.g., January, October)")
15     travel_style: Optional[str] = Field(
16         default="balanced",
17         description="Travel style: adventure, relaxation, cultural, balanced"
18     )
19     budget: Optional[str] = Field(
20         default="mid-range",
21         description="Budget level: budget, mid-range, luxury"
22     )
23
24
25 class TravelFlowOutput(BaseModel):
26     """Output schema for the India travel planning flow."""
27     summary: str = Field(description="Complete travel plan summary including weather, itinerary,")
28
29
30 # -- Helper Tools (self-contained) --
31
32 ITINERARY_QUICK = {
33     "goa": "Day 1: Calangute Beach & water sports → Tito's Street dinner. Day 2: Old Goa churches",
34     "jaipur": "Day 1: Amber Fort & Nahargarh Fort → Johri Bazaar. Day 2: City Palace & Hawa Mahal",
35     "kerala": "Day 1: Fort Kochi heritage walk → Kathakali show. Day 2: Alleppey houseboat cruise",
36     "manali": "Day 1: Old Manali & Hadimba Temple. Day 2: Solang Valley adventure. Day 3: Rohtang"
```

The interface includes a sidebar with the Explorer view showing the project structure: `WATSONX ORCHESTRATE` > `Agents` > `Tools` > `Connections` > `Knowledge Bases` > `Toolkits`. The Environment Manager at the bottom shows the current environment and options to activate or add new environments.

On the right, the IBM BOB panel shows the "Create travel..." task with a progress bar and a list of files being created: `travel_agent/tools/india_destinations.py`, `travel_agent/tools/travel_planner.py`, and `travel_agent/tools/weather_info.py`. The "Tools awaiting approval:" section shows a "Write File ..." button and an "Approve once" button.

IBM Bob Screenshots:



File Edit Selection View Go Run Terminal Help

← → bob test

Upgrade your plan

EXPLORER

BOB TEST

.venv

static

script.js

style.css

templates

index.html

.env.example

.gitignore

app.py

DEPLOYMENT.md

Procfile

README.md

requirements.txt

runtime.txt

SETUP_GUIDE.md

requirements.txt

.env.example

app.py

index.html

script.js

README

CODEX

IBM BOB

def get_watsonx_model

73 # =====

74 # IBM Watsonx.ai Configuration

75 # =====

76

77 def get_watsonx_model() -> Model | None:

78 """Initialize and return IBM Watsonx.ai model"""

79 try:

80 credentials: Credentials = Credentials(

81 url=os.getenv('IBM_WATSONX_URL', 'https://au-syd.ml.cloud.ibm.com'),

82 api_key=os.getenv('IBM_CLOUD_API')

83)

84

85 project_id: str | None = os.getenv('IBM_WATSONX_PROJECT_ID')

86

87 model: Model = Model(

88 model_id='ibm/granite-guardian-3-8b',

89 params=

90 GenParams.DECODING_METHOD: "greedy",

91 GenParams.MAX_NEW_TOKENS: 900,

92 GenParams.MIN_NEW_TOKENS: 50,

93 GenParams.TEMPERATURE: 0.7,

94 GenParams.TOP_K: 50,

95 GenParams.TOP_P: 1

96),

97 credentials=credentials,

98 project_id=project_id

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

BOB FINDINGS

requests==2.31.0

- urllib3==2.7.0

+ urllib3==2.1.0

+ werkzeug==3.0.1

+ zipp==4.1.0

warning: 'ibm-watson-machine-learning==1.0.335' is yanked (reason: "Causes errors on py 3.9")

(.venv) PS D:\OneDrive - Edunet Foundation\Desktop\bob test>

Create an AI-powered Nutrition ... 42x / 200k 1.17

All tasks completed! 13/13

1. Install dependencies: pip install -r requirements.txt

2. Copy .env.example to .env and add IBM credentials

3. Run: python app.py

4. Open: http://localhost:5000

Customization:

The AGENT_INSTRUCTIONS section in app.py allows you to easily customize:

- Agent tone and personality
- Diet specializations
- Indian food preferences
- Safety rules and limitations
- Response formatting

All files are production-ready with comprehensive error handling, security best practices, and deployment configurations for Heroku, IBM Cloud, AWS, and Docker!

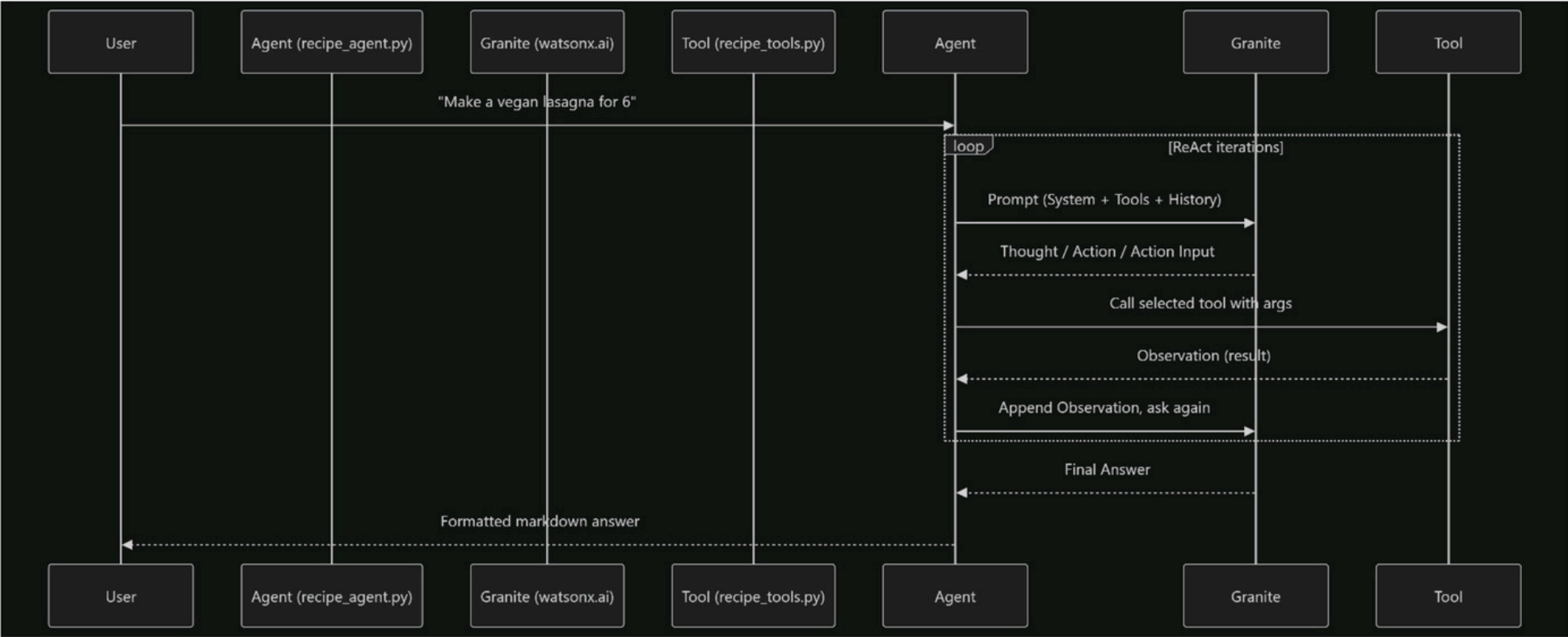
Start New Task

What's on your mind?

(@ to add context, / for commands, hold shift to drag in files/images)

Code

Architecture blueprint for IBM bob:



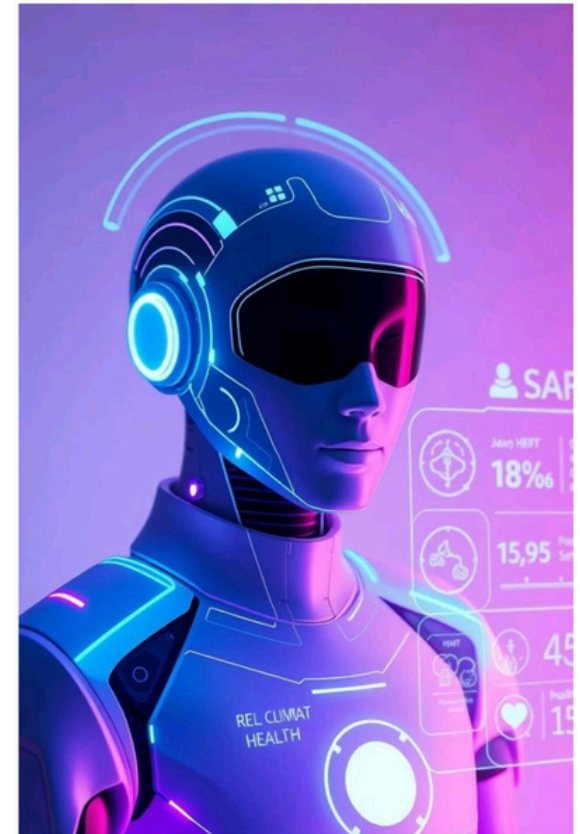
Role of Agentic AI in the solution

Role of Agentic AI in Nutrition Agent

Agentic AI enables the Nutrition Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized nutrition guidance. Each agent performs a specific role—such as retrieving nutritional data, generating diet plans, providing health advice, and analyzing food logs—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate results. Agentic AI also brings **context awareness**, understanding user details like age, health conditions, and fitness goals to generate tailored recommendations. It supports **real-time adaptation**, continuously updating suggestions based on user inputs like daily meals.

Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

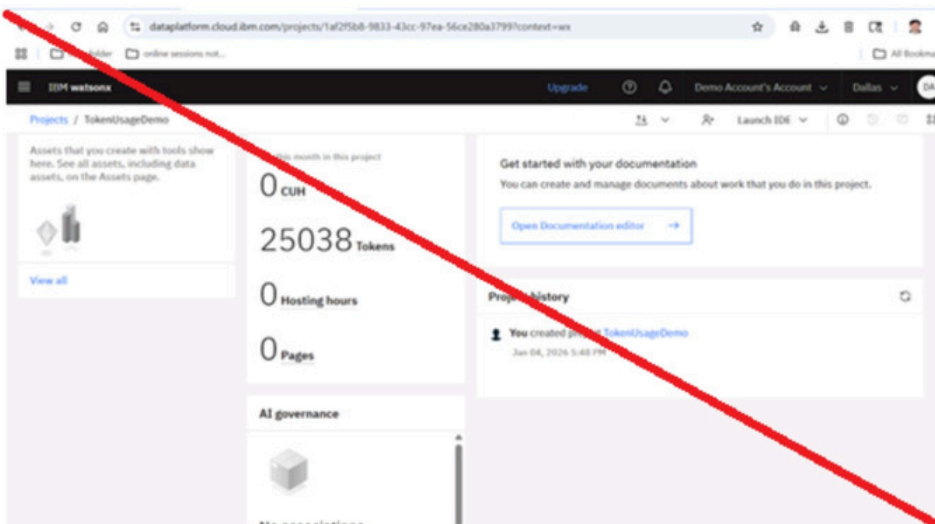
Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital nutritionist** that helps users maintain a healthy lifestyle.



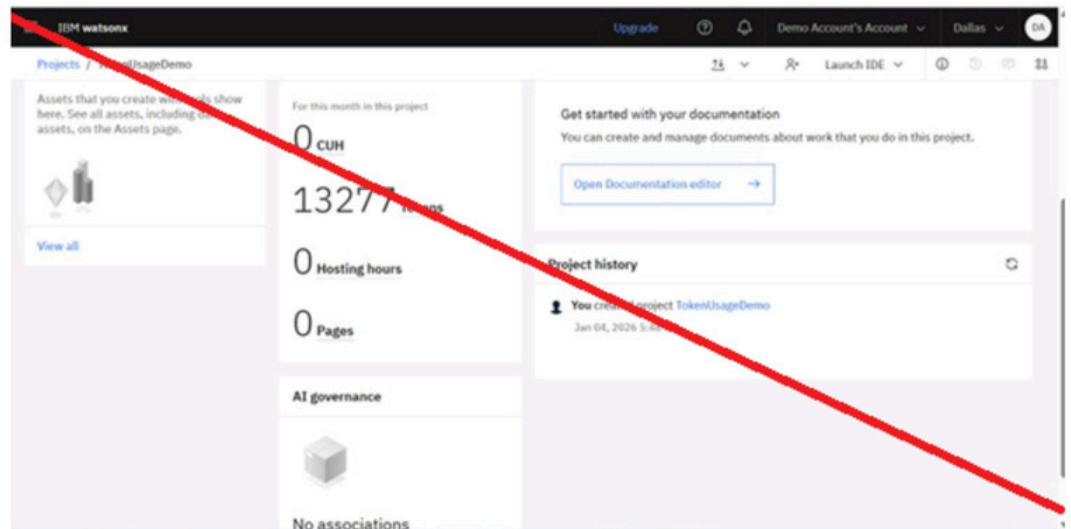
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Your Personal AI Nutrition Expert

Get personalized nutrition advice, meal plans, and health insights powered by IBM watsonx.ai and Granite models. Specialized in Indian cuisine and family nutrition.

AI Advice

Powered by IBM Granite models.

Indian Cuisine

Specialized in regional diets.

Family Plans

Customized health profiles.

Chat with NutriBot

Ask anything about nutrition, diet, and healthy eating

Welcome to NutriBot!

I'm your AI nutrition expert. Ask me about:

- Personalized meal plans
- Calorie counting & macros
- Indian food nutrition
- Weight loss or muscle gain diets

Chat with NutriBot

Ask anything about nutrition, diet, and healthy eating

Welcome to NutriBot!

I'm your AI nutrition expert. Ask me about:

- Personalized meal plans
- Calorie counting & macros
- Indian food nutrition
- Weight loss or muscle gain diets

Create one day diet plan for me name rajesh age 25 height 5.8 weight 70 gender male goi

Send

Get personalized nutrition advice, meal plans, and health insights powered by IBM watsonx.ai and Granite models. Specialized in Indian cuisine and family nutrition.

AI Advice

Powered by IBM Granite models.

Indian Cuisine

Specialized in regional diets.

Family Plans

Customized health profiles.

Chat with NutriBot

Ask anything about nutrition, diet, and healthy eating

Sure, here is a one-day diet plan for Rajesh:

Breakfast:

- 1 cup of oats with 1/2 cup of milk and 1/2 cup of yogurt
- 1/2 cup of mixed berries
- 1/2 cup of chopped nuts

Lunch:

- 1 cup of brown rice or 2 rotis
- 1/2 cup of mixed vegetables
- 1/2 cup of chickpeas or dal
- 150g grilled chicken breast

Dinner:

- 1 bowl vegetable soup
- 2 boiled egg whites with steamed veggies

Novelty and Uniqueness

The proposed Nutrition Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.

RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy recommendations.

Hyper-Personalization – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on daily food logs and user behavior.

Multi-Modal Input – Supports text, voice, and image-based meal tracking for better usability.

Preventive Healthcare Focus – Goes beyond diet planning to provide disease-specific nutrition advice.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Submission Details

- Student: Satyam Kumar
- Student ID: STU68f7c371dc3df1761067889
- Email: gamingytsk182@gmail.com
- WhatsApp: 8092033232
- GitHub: <https://github.com/DevSatyam7>
- Project: Nutrition Agent – Problem Statement No. 8

Verify that the GitHub repository is accessible and contains the relevant project files/code.

Selected problem statement and technology requirement are taken from the provided AICTE-2026 document.

Future Scope

1. Integration with Wearables & Health Devices –

The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the Nutrition Agent to provide more accurate and dynamic diet recommendations based on actual user activity and health metrics.

2. AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Users can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse populations, especially in regions with varied linguistic backgrounds.

In recognition of the commitment to achieve
professional excellence



Satyam Kumar

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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This certificate is presented to

Satyam Kumar

for the completion of

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According to the Your Learning Builder - Plans system of record

Thank You!

Thank you for your time and interest.